CHAROTARUNIVERSITY OF SCIENCE & TECHNOLOGY
4^{th} Semester of B. Pharm. Examination
University Theory Examination November 2016
PH212 Physical Pharmaceutics

Date: 09.11.2016, Wednesday **Time:1:30 p.m to 4:30 p.m** **Maximum Marks: 80**

Instructions:
1. **Figures to right indicate marks.**
2. **Section wise separate answer book to be used.**
3. **Draw neat sketches wherever necessary.**

SECTION- I

Q 1	Attempt any **TWO** of the following	
A	(i) Enlist the methods for particle size determination. Briefly explain any one method.	**05**
	(ii) Discuss the derived properties of powder.	**05**
B	Write a detail note on 'Non Newtonian fluids' with examples.	**10**
C	What is meant by controlled flocculation? Discuss the various means by which controlled flocculation can be achieved.	**10**
Q 2	Attempt any **TWO** of the following	
A	Discuss about the various signs indicating the instability of emulsion formulation.	**10**
B	(i) Define: Rheology. Give the application of rheology in pharmacy.	**05**
	(ii) Classify various instruments used for measurement of viscosity. Describe any one viscometer to find out viscosity of Newtonian fluids.	**05**
C	Define Angle of repose. Describe angle of repose and write relationship with powder flow.	**10**

SECTION- II

Q 3	Attempt any **TWO** of the following	
A	Define : Surface tension. Describe the 'Capillary rise method' as a method for the determination of surface tension.	**10**
B	Write a short note on:	**10**
	(i) Zeta Potential (ii) Spreading co-efficient	

C Explain the following in brief: **10**

(i) Differential scanning calorimetry (ii) X-ray Diffraction

Q 4 Attempt any **TWO** of the following

A Define: Complex compounds. Classify 'Complexes' and describe the application of complexation in pharmacy. **10**

B Write a detailed note on: Polymorphism. **10**

C Define: Buffer capacity. Write a detailed note on any physiological buffer system. **10**
